
Hidden-Feature Matching and Left-Right Movement in Gait Representations

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Abstract

Temporal representations should retain information that helps describe an evolving system, alongside the features they learn to predict. We examine this requirement in gait using a small skeleton Joint-Embedding Predictive Architecture (JEPA). The model predicts hidden skeleton features; a separate regression estimates a signed contrast between left and right movement speeds from its frozen encoder. On 625 clips from 93 held-out-by-source videos, trained predictors more consistently match hidden features from the correct clip than from another video. Yet with an expanded feature summary, mean source-balanced R^2 for the movement estimate falls from 0.22 at initialization to 0.10–0.11 after training. Motion-weighted and connected-region masks do not establish an improvement over their own random references. This focused negative finding connects latent prediction to evaluation of observable movement: success on hidden-feature matching did not improve the specified movement estimate under the tested training and readout procedure. Future-only prediction remains a separate test.

1 Introduction

Walking is an evolving system: coupled body landmarks change position over time, and movement depends on their trajectories. Rhythm, phase and coordination make gait an appealing setting for temporal modeling, although video-derived coordinates describe only part of the physical process. They omit forces and contact and are affected by camera view and pose-estimation error. Temporal world modeling aims to represent such a system well enough to predict its observable behavior. A practical evaluation question is whether a representation trained through prediction preserves information needed for a specified movement measurement.

We study that question with S-JEPA [1], using a left–right contrast of median coordinate speeds. The contrast is clinically motivated by work on gait asymmetry [2], but is not a validated clinical index. Our central null is that JEPA training provides no improvement over the same encoder at initialization when a frozen readout estimates this contrast on held-out source videos. We investigate input preparation, hidden-feature correspondence, masking, and movement readout as distinct parts of that question. The current expanded-readout and mask comparisons are exploratory analyses on a repeatedly examined cohort; they do not retrospectively become registered tests.

Predictive skeleton methods and temporal foundation models share an evaluation problem: the training target is a learned feature, whose successful prediction need not preserve every observable of interest. Our contribution is a controlled, domain-specific test of this issue. It is relevant to evaluation and reliability, grounded in temporal data and a temporal encoder, without establishing foundation-model scale, broad transfer, persistent state or a gait simulator.

2 Study design

Data and observable. Our selected GAVD subset [3] contains 666 annotations, 642 matching pose archives and, after quality checks, 625 clips from 93 source videos. Each archive has shape $T_i \times 33 \times 4$: three estimated coordinates plus landmark visibility, with original frame numbers and frame rate. Coordinate units are image normalized with inferred depth; they are not calibrated physical speeds.

The movement path uses observed coordinates, pelvis centering and a body-width scale, without gap filling or resizing. For five pairs—shoulders, knees, ankles, heels and foot tips—we calculate median speeds $m_{L,k}, m_{R,k}$ using original timestamp differences. Both sides must be observed at both ends of at least eight common transitions for every pair. The signed target is

$$y = \frac{1}{5} \sum_{k=1}^5 \frac{m_{L,k} - m_{R,k}}{m_{L,k} + m_{R,k} + 10^{-8}}.$$

Anatomical reflection exchanges left/right landmarks and validity flags and flips the centered horizontal coordinate, giving $y(Mx) = -y(x)$. Hips define the pelvis and are excluded from the contrast because centering equalizes their speed magnitudes. The statistic does not measure phase, stride timing or loading; reversing a trajectory together with its sampling intervals preserves its speed magnitudes.

Encoder input. A landmark is observed when visibility is at least 0.45 and all coordinates are finite. A separate encoder path fills interior gaps of at most four samples, treating filled positions as valid. After pelvis centering and body-width normalization, coordinates and validity are linearly resized to 64 positions by sequence index. Original time intervals are discarded. Resized validity must reach 0.999; invalid coordinates are zero. Each joint’s four-position patch projects 12 coordinates to 96 features, giving a 16×33 grid. A patch is valid only if all four positions are valid. Anatomical identities and prepared positions survive this operation; the original trajectory, duration and cadence need not.

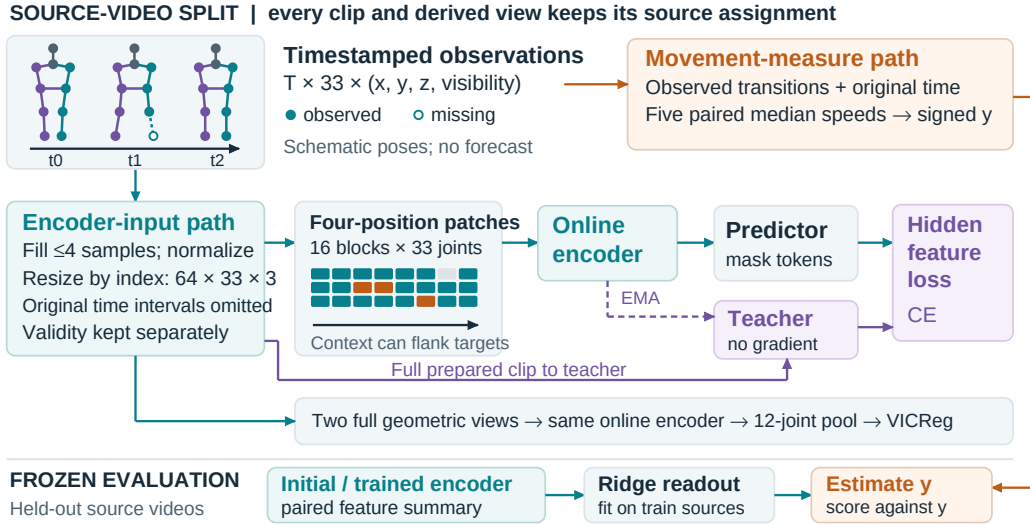


Figure 1: Two measurement paths and the completed JEPA training grid. Poses and mask cells are schematic. The movement target reaches only the separate regression evaluation. The online encoder has visible context on either side of a hidden position; the teacher and full-view regularizer access complete prepared clips.

Training. The completed grid uses 96 features, encoder/predictor depths 4/2, four attention heads, batch size 20 and 1,200 AdamW updates, with learning rate 0.001 and weight decay 0.05. The online encoder zeroes hidden patch content before adding joint and position embeddings; the predictor substitutes learned mask tokens at target positions. Cross-entropy matches centered teacher feature

distributions at hidden valid positions, averaged within clips and then across clips. The teacher has no gradients and follows the online weights by exponential moving average (momentum 0.999). A full-view VICReg term [4], weighted 0.05, pools twelve gait joints across two geometric views. Thus the total objective also accesses complete clips. Training uses CUDA BF16 and evaluation FP32.

We compare three motion conditions and two region conditions across five folds and five seeds: 125 encoders. The motion conditions use random sampling, a MAMP-style sampler [5], and a robust motion mixture. Their target count is half the batch’s smallest valid twelve-joint gait pool, rounded down, sampled across all 33 joints. Connected regions hide six connected landmarks over eight of sixteen blocks, with a separate count-matched random reference. Starting weights, source draws, geometric views and update counts are matched within each comparison. No reflection augmentation or reflection loss appears in this grid.

Held-out readout. Five outer folds keep each video’s clips and derived views together. Fresh encoders fit only outer-training videos with source-balanced sampling. All overlapping windows, mirrors and augmentations inherit the source split. The encoder is frozen while ridge regression, a linear model with a coefficient penalty, estimates y . Three inner source folds select the penalty using only outer-training labels; imputation and scaling are fitted within each inner training set and then refitted with the regression on all outer-training sources. Inner-validation clips may have contributed without labels to outer-training encoder fitting.

The compact readout uses left-minus-right and left-plus-right mean features from five pairs (960 inputs). The expanded readout adds feature standard deviations, absolute changes between adjacent valid blocks, and ten support fractions (2,890 inputs). Both initial and trained encoders use this procedure. We pool outer-test predictions within each seed, weighting each video equally, then average the five R^2 values. Paired 95% source-bootstrap intervals resample videos with their clips and seed predictions together (2,000 resamples). They condition on the fitted models and existing splits. Repeated seeds and mask draws do not add independent sources.

3 Findings

Movement readout falls after training. The expanded initial representation reaches mean $R^2 = 0.223$, compared with 0.101–0.114 for trained teachers (Figure 2). All five conditions decrease in every seed, with greater mean absolute error. Paired trained-minus-initial intervals are below zero in every condition. For the random motion mask, the difference is -0.109 with interval $[-0.170, -0.041]$. Final online encoders also score below their matched initial encoders.

The compact initial readout scores 0.071; expanding it increases R^2 by 0.152. The added components mix variation, feature changes and observation support, so the gain cannot be assigned specifically to temporal order. The initial control retains the pretrained pose detector, input preparation, anatomical identities and supervised regression. A 264-value direct-pose summary reaches $R^2 = 0.035$; this limited baseline does not bound stronger coordinate readouts.

Correct-clip features become easier to match. Each diagnostic compares source-weighted mean squared feature error against the correct clip and valid targets from another source, holding the evaluated clips and hidden positions fixed. All 375 trained checks favor correct targets; 33 of 75 initial checks do so. The denominator is models times evaluation masks, not successful predictions of 375 independent movements. This preference could involve pose, viewpoint or missingness as well as motion. Teachers and target distributions change during training, so raw training losses cannot rank representation quality across conditions.

Changing mask geometry does not establish better readout. The inspection draws confirm that motion samplers select higher-motion targets and that connected regions remove immediate temporal brackets. Motion masks hide approximately 17% of valid all-landmark tokens; regions hide 9.9%. Their readout comparisons therefore use distinct random references. MAMP-style, motion-mixture and region differences are respectively -0.0008 , $+0.0005$ and -0.009 , with intervals $[-0.020, 0.015]$, $[-0.015, 0.015]$ and $[-0.033, 0.018]$. Intervals spanning zero leave benefits uncertain, rather than establishing equivalence.

Preparing input changes the measurement. A saved comparison recalculates the target after input preparation: 623 clips from 92 sources have finite values on both paths, with source-weighted sign

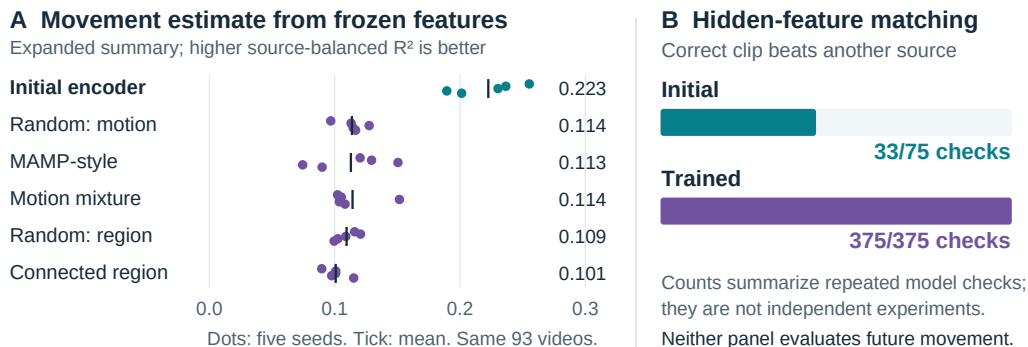


Figure 2: Different outcomes from the same fitted models. A: frozen movement readout; points are five seed scores, and short vertical marks their mean. B: correct-clip feature error is lower in 375 of 375 trained diagnostic checks, versus 33 of 75 distinct initial checks. These checks reuse models and videos.

115 agreement 70.4% and calculation-agreement $R^2 = 0.218$. The two paths disagree, but this calculation
116 neither sets a model-performance ceiling nor identifies an individual preprocessing cause.

117 4 Interpretation and next experiment

118 The two evaluations answer different questions. JEPA training improves preference for the correct
119 clip’s hidden features, while a fixed regression procedure estimates its left–right speed contrast less
120 accurately. Poorer readout performance does not imply complete destruction of movement information.
121 In particular, 49/125 teacher fits and 96/125 online fits select the largest ridge dimension penalty, 10,000.
122 Wider training-only selection, support-only controls and summaries of matched dimension would
123 help separate feature geometry from readout limitations. Person identities, reuploads, pose errors
124 and recording cues remain possible explanations beyond what source holdout controls.

125 A next experiment would predict future coordinates from a past-only prefix at common valid end-
126 points and physical-time horizons, against last-position, constant-velocity, direct-pose and initial-
127 encoder controls. Any decoder and settings must be fitted on training sources only. Correctly paired
128 training futures should be compared with mismatched futures. Future coordinates or visibility must
129 not affect prefix preparation, normalization, mask selection or forecasts. Existing past-feature read-
130 out and future-feature decoding demonstrations have synthetic training evidence; a real-data horizon
131 census establishes eligibility only. Decoding observed future-teacher features diagnoses the decoder;
132 decoding predictor features produced from the past tests forecasting. Multiple horizons from one
133 prefix are not recursive rollout evidence.

134 This study supports evaluating latent prediction alongside access to specified observable behavior.
135 The evidence remains conditional on one cohort, JEPA recipe and readout, with no participant-
136 independent or distribution-shift evaluation. Forecasting and simulation would demand stronger
137 tests of future behavior and consistency than this whole-clip contrast supplies.

138 5 Related work

139 S-JEPA evaluates hidden-feature prediction through skeleton action recognition [1], whereas MAMP
140 reconstructs motion and uses motion-aware masking [5]. Connected-body masking in SLiM aims
141 to reduce local shortcuts [6]. Our controlled masks borrow these ideas while holding the JEPA
142 objective fixed; they do not reproduce those complete methods. V-JEPA 2 couples video representa-
143 tions with anticipation and action-conditioned planning [7], and seq-JEPA separates transformation-
144 sensitive from invariant features [8]. Together these works motivate evaluation beyond latent loss.
145 Our narrower contribution compares matched initial and trained features on an observed gait quan-
146 tity, without claiming their forecasting or world-modeling capabilities.

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A Measurement and implementation details

The archive coordinates are $x = (x_0 + uW_c)/W$, $y = (y_0 + vH_c)/H$, and $z = dW_c/W$, where crop origin and size are (x_0, y_0) , W_c, H_c , frame size is W, H , and u, v, d are pose estimates. Unequal image-axis scaling and inferred depth remain after body normalization. The scale is the median of valid shoulder and hip widths. The movement path requires observed hips; the encoder uses the median valid pelvis midpoint for missing hips (zero if none is available). No boundary extrapolation fills gaps.

The teacher receives the unaugmented prepared clip; the masked online path receives a geometric view. Two full online views use rotations up to 8 degrees in the horizontal-depth plane and translations up to 0.03 in image-plane coordinates. Target/predictor temperatures are 0.06/0.10; the teacher center uses momentum 0.9. Gradient norm is clipped at 1. The twelve-joint regularizer pool is shoulders, hips, knees, ankles, heels and foot tips. Its projector is a separate training component. It combines invariance, variance and covariance penalties with internal weights 25, 25 and 1, then overall weight 0.05.

The expanded summary uses common left/right validity within each pair; increments require consecutive common valid blocks. Each pair contributes differences and sums of means, standard deviations and mean absolute feature increments (six 96-vectors), plus two support fractions. Thus $5 \times 6 \times 96 + 10 = 2,890$. Missing support gives zero summary values. Temporal positions in this summary are prepared block indices, not original timestamps. Ridge penalties are selected from 0.01, 0.1, 1, 10, 100, 1000, 10000.

Source-balanced R^2 uses weights $w_{vi} = 1/(Vn_v)$ and weighted test mean $\bar{y}_w$:

$$R_w^2 = 1 - \frac{\sum_{vi} w_{vi} (y_{vi} - \hat{y}_{vi})^2}{\sum_{vi} w_{vi} (y_{vi} - \bar{y}_w)^2}.$$

The test mean defines this metric’s denominator, not a deployable fitted baseline. Outer test folds contain 19, 19, 19, 18 and 18 videos and 189, 182, 72, 77 and 105 clips. Source IDs and split membership are checked; no persistent participant identities or complete cross-upload content comparison establish independence across different IDs. Annotation retrieval/version records are incomplete.

B Supporting results and reproducibility

Training condition	Mean R^2	Trained minus initial	95% paired source interval
Random, motion budget	0.114	-0.109	[-0.170, -0.041]
MAMP-style	0.113	-0.110	[-0.165, -0.051]
Motion mixture	0.114	-0.108	[-0.171, -0.042]
Random, region budget	0.109	-0.113	[-0.166, -0.060]
Connected region	0.101	-0.122	[-0.183, -0.055]

These exploratory percentile intervals use bootstrap seed 812, retain all five fitted seeds and have no familywise or development-selection adjustment. The numerical supplement preserves aggregate seed scores and recorded intervals. The executed report identifies 125,000 prediction rows and 200 pooled score rows; a saved audit records their recomputation. Raw predictions and checkpoints are unavailable in the revision environment, so this revision checks aggregate consistency without claiming to repeat that bootstrap or inference.

A separate reflection-augmentation comparison has real-data summary evidence: augmentation reduces the specified teacher-token reflection discrepancy by $\Delta q = -0.008$ $[-0.010, -0.007]$, while its readout difference is $\Delta R^2 = 0.004$ $[-0.006, 0.013]$. This uses its own recipe, four inner readout folds and compact summary; it is not pooled with the motion grid. An explicit reflection loss has

217 synthetic evidence only. Likewise, a readout forced to take the form $g(x) = [h(x) - h(Mx)]/2$
218 reverses sign algebraically without establishing learned movement information.

219 GAVD annotations describe the cohort, not affected side or treatment response. Uncalibrated coordi-
220 nates and absent clinical reference measurements preclude a clinical interpretation of y . Public
221 annotation access does not authorize redistribution of separately hosted videos or derived poses;
222 neither is included in the supplement.

223 **AI assistance.** An AI assistant supported author-side evidence reconciliation, literature checking,
224 manuscript revision, figure programming and adversarial checks. No new model training was per-
225 formed for this revision. Responsibility for the submitted text, figures, claims and references remains
226 with the authors.